HyCSA: Finding Overlapping Hyperbolic Communities

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Abstract

A classical model for communities in a social network is a collection of non-overlapping quasi-cliques, such as created by the stochastic block models. This is very restricted model, and many approaches have been proposed to address these shortcomings. In this paper, we present a method for finding hyperbolic communities. Hyperbolic communities are a powerful model for communities in social networks. They generalize stars, cliques, and power-law communities while being easy to interpret, making them an idea choice for modeling real-world communities. Our algorithm can find these communities in real-world social networks even if the communities overlap moderately. This is the first known method for finding such communities and we show that our algorithm can find the communities which are more closely the the ground-truth communities than communities found by competing methods.

Keywords

Community detection, Simmulated annealing, Random walks

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1 Introduction

Community detection has long been one of the key problems in social network analysis. Traditionally, communities have been modelled as (quasi-) cliques, that is, subgraphs where (almost) every member knows (almost) every other member. Such models, however, are long known to be limited and not very realistic, so-called "caveman communities" (see, e.g. [12]).

In this paper, we concentrate on *hyperbolic communities* [13], a modern definition of a community that splits the community membership into clique-like *core members* and peripheral *tail members* (formal definitions are provided in Subsection 2.1). Hyperbolic model, like other similar models (e.g. [4, 6]), distinguishes between *vertex membership* and *edge membership* in a community: two vertices, that are neighbors to each other, can be members of the same community even if the edge connecting them is not considered part of the community.

Our main contribution is an algorithm to find overlapping hyperbolic communities from given social network. While hyperbolic communities are found to fit well to real-world data [13, 14], so far there has not been a way to find them. Instead, the existing work has fitted the model to known ground-truth communities, that is, it has assumed that the vertex memberships in the communities are known, and then it has identified the edge memberships. In contrast, our algorithm can find the vertex and edge memberships, given only the number of communities to search. Furthermore, our algorithm can find communities that are overlapping, and it can identify vertices that do not belong to any community.

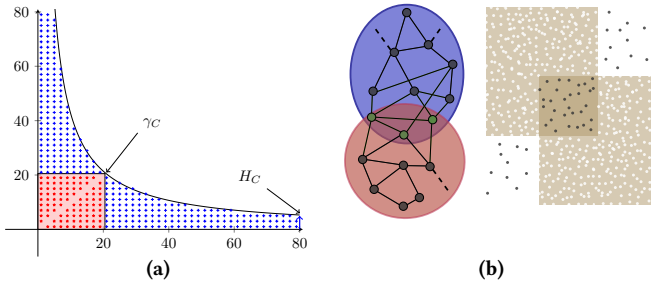


Figure 1: (a): Schematic view of a hyperbolic community. x and y axes correspond to vertices and edges between them are locations (x, y). Red stars indicate edges that are in the core and blue plus-signs edges that are in the tail. The size of the core is γ_C and the height of the tail is H_C. (b): Two overlapping communities and corresponding adjacency matrix.

2 Background and Related Work

2.1 Definitions

In this paper, we consider undirected graphs $G = (V, E)$. If $C \subseteq V$, we write $G|_C$ for the subgraph of G induced by C, i.e. $G|_C = (C, E \cap C \times C)$. A *core-tail community* [13] in graph G is a tuple $(C, \pi_c, \gamma_C, H_C)$, where $C \subseteq V$ is the set of vertices in the community, $\pi_C \colon C \to \{0, 1, \ldots, |C| - 1\}$ is a permutation of C to integers in descending induced degree order (i.e. vertex 0 is the vertex with the highest degree in $G|_C$), and $\gamma_C \geq 0$ and $0 \leq H_C \leq \gamma_C$ are parameters for the "shape" of the community. In particular, denote

$$p_C = \frac{\gamma_C^2 - (|C| - 1)H_C}{(|C| - 1) + H - 2\gamma_C} \quad \text{and} \quad \Theta_C = \frac{(\gamma_C - H_C)^2(\gamma_C - |C| - 1)^2}{(|C| - 1 + H_C - 2\gamma_C)^2}, \tag{1}$$

and let $i \neq j$ be indices of vertices in C per π_C. Then edge $\{i, j\}$ is *in the community* if and only if

$$(i + p_C)(j + p_C) \leq \Theta_C. \tag{2}$$

Equation (2) should make the origin of the name *hyperbolic community* clear. The core-tail model (called `fixed` in [13]) has easier interpretation, though: parameter γ_C is the size of the *core*, that is, the clique-like part, of the community, and parameter H_C is the height of the *tail* at the end of the community. Figure 1a illustrate an example of a hyperbolic community. Every community is associated with two sets of edges: the area of the community, A_C, defined as $A_C = \{(i, j) \in C \times C : f(i, j, \theta) = 1\}$, and the edges of

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it $E_C = E \cap A_C$. We can also define their complements as follows: $\overline{A_C} = (C \times C) \setminus A_C$ and $\overline{E_C} = A_C \setminus E$.

2.2 Related Work

Community detection is a fundamental problem in graph analysis, traditionally focusing on detecting densely connected groups of vertices, such as quasi-cliques [3]. These networks are well studied using stochastic block models [1, 2]. However, communities in real-world networks frequently do overlapping [11, 17]. This has shifted the research toward finding overlapping communities, including clique percolation [15], modularity-based methods [5, 16], spectral methods [11], fuzzy clustering [18], information theory based methods [4, 8], and matrix factorizations [9, 17].

On the other hand, there has been an interest in finding non-clique communities, such as the core–periphery model [6]. More recently, "smoother" boundary between core and non-core vertices has gained popularity, including the HyCoM-FIT (HyCom) model [4] and the hyperbolic community model [13]. The latter is shown to fit well to real-world graphs [9, 13, 14]. For comparison between these models, we refer the reader to [13].

3 The HyCSA Algorithm

The HyCSA algorithm (Hyperbolic Communities via Simulated Annealing) has two steps: in the first step, it finds seed communities, while in the second step it starts expanding the seed communities using simulated annealing. The seeding process is described in Section 3.2 while the expansion is described below.

3.1 Expanding Seed Communities

The expansion method can both add and remove vertices. The seed communities are not necessarily the final cores of the communities; some vertices in the seed can even be removed from the final community. The expansion method is based on standard simulated annealing approach, where the algorithm moves to a neighboring solution (adds or removes a vertex) randomly, depending on how good the solution is and the temperature in the system.

Let C' denote a set of proposed community nodes obtained by either adding or removing a vertex from C. Following the likelihood formulation in (15) from [13], we evaluate the quality of a candidate update by considering only the likelihood terms affected by the modification of the current community. The corresponding objective is defined as

$$\ell(C' \mid C) = |E_{C'}| \log(d_{C'}) + |\overline{E_{C'}}| \log(1 - d_{C'}) + |E \setminus A_{C'}| \log(d'_O) + \left(|\overline{A_{C'}}| - |E \setminus A_{C'}|\right) \log(1 - d'_O), \quad (3)$$

where $A_{C'} = A_{C'} \cup \bigcup_{D \in C,\, D \neq C} A_D$ denotes the union of the community areas after applying the proposed update and C is set of predicted community nodes. The global outside density is recomputed for each proposed state as $d'_O = \frac{|E \setminus A_{C'}|}{|\overline{A_{C'}}|}$.

The quality of a proposed move is measured by the change in the objective, $\Delta = \ell(C' \mid C) - \ell(C \mid C)$. A proposal that improves the likelihood, i.e., $\Delta \geq 0$, is always accepted. Otherwise, it is accepted according to the simulated-annealing probability $P_{\text{acc}} = \exp(\frac{\Delta}{T}), \Delta < 0$, where T denotes the temperature. Following the standard simulated annealing process, the temperature is lowered after every update step. This probabilistic acceptance of non-improving moves allows the search to escape poor local optima during community expansion. Full pseudo-code of the algorithm is provided in the online appendix.[1]

3.2 Seeding Methods

We utilize following initialization methods in our experiments to compute initial communities for HyCSA. They are referred as $\text{HyCSA}_{\text{rss}}$ and HyCSA_{rw}, respectively.

3.2.1 Representative Seed Selection (rss): This seeding method is used to evaluate how well the expansion process works when it gets a known-good seed. It creates the seed by sampling vertices from ground-truth communities according to the vertices' within-community degree. This process gives a representative sample for each community, but it is obviously not realistic seeding method, as it requires a priori knowledge of the communities.

3.2.2 Random Walks (rw): Our main seeding method is based on random walks to find densely connected subgraphs (i.e. potential cores of communities). We perform multiple independent random walks of fixed maximum length. Each walk begins from the same starting vertex, and at every step, the next vertex is selected uniformly at random from the neighbors of the current vertex. During the walks, we record the number of times each vertex is visited. Vertices that are visited frequently are considered more strongly associated with the local neighborhood of the starting vertex.

To this end, let $c(v)$ denote the total number of visits to vertex v across all random walks. After completing the walks, the visited vertices are ranked in descending order according to $c(v)$. We retain the top $k = \max\{1, \lceil \rho_{\text{rw}} L \rceil\}$ vertices, where L is the random walked length and ρ_{rw} is a user defined selection percentage. The resulting set of highly visited vertices forms the candidate seed used for subsequent community initialization.

The random walk process is repeated $k \log k$ times, where k is the user-supplied parameter for the number of communities. Following the Coupon Collector's lemma, this should give us a good probability of having at least one starting vertex in every community, assuming that there are k true communities.

To obtain our final k community seeds, we employ a greedy selection strategy based on Jaccard similarity. First, one seed is selected uniformly at random to initialize the selected set. For each remaining candidate seed S, we compute its Jaccard similarity with every previously selected seed T, $J(S,T) = \frac{|S \cap T|}{|S \cup T|}$.

The score of a candidate is defined as its maximum Jaccard similarity with any already selected seed, $\phi(S) = \max_{T \in \mathcal{S}} J(S,T)$, where $\mathcal{S}$ denotes the current set of selected seeds. At each iteration, we select the candidate with the smallest value of $\phi(S)$, $S^* = \arg\min_{S \in \mathcal{R}} \max_{T \in \mathcal{S}} J(S,T)$, where $\mathcal{R}$ is the set of remaining candidates. The selected seed is then added to $\mathcal{S}$, and the procedure is repeated until the desired number of seeds is obtained. This criterion favors seeds whose largest overlap with the previously selected seeds is as small as possible. Consequently, the selected seeds are encouraged to represent distinct regions of the graph, reducing redundancy among the initial community seeds.

[1] https://anonymous.4open.science/r/HyCSA-Finding-Overlapping-Hyperbolic-Communities-D24F/

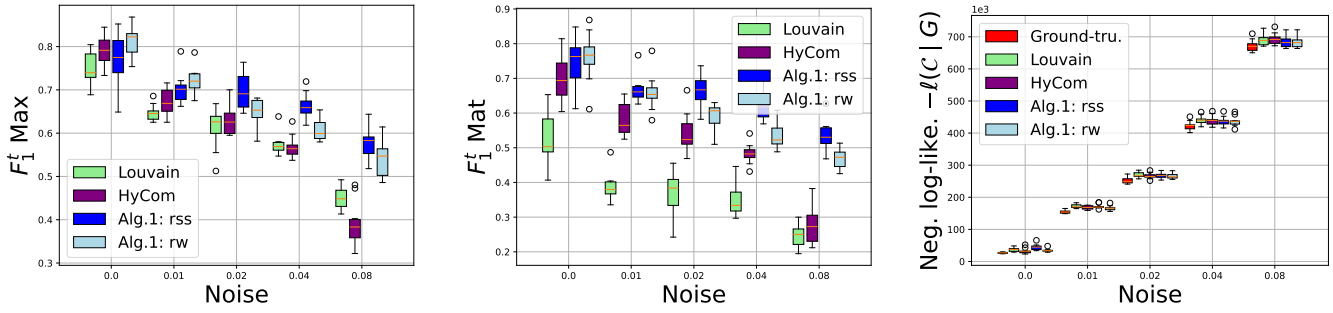


Figure 2: Synthetic data with overlapping communities.

4 Experimental Evaluation

We evaluate the performance of HyCSA on both synthetic and real-world networks. The aim of the synthetic data experiment is to study its performance under varying levels of noise on synthetic overlapping communities. Next we evaluate it with real-world networks. Our implementation is available from the aforementioned anonymous repository.

4.1 Baselines

We compare the performance of HyCSA against two established community detection methods: Louvain [5] and HyCom [4]. The Louvain algorithm is a widely used community detection method based on modularity optimization [5]. For our experiments, we use its implementation provided by the NetworkX library [7]. In contrast, HyCom is based on the minimum description length principle and captures the hyperbolic structures in the real-world networks [4]. These two methods provide benchmarks for evaluating our proposed algorithm.

4.2 Measures

We measure the results of the algorithms using the log-likelihood of the data given the model and by measuring the similarities between the found and ground-truth communities. Not all methods find the same number of communities, which creates challenges for fair comparison. To that end, we use two measures, F_1^t Max and F_1^t Mat. They both compute the average F_1 score between the true and found communities, but they match the communities differently.

For F_1^t Max, each ground-truth community is matched to the found community that yields the best F_1 and the average is taken over the ground-truth communities. For F_1^t Mat, a matching is first computed between the ground-truth and found communities. The matching maximizes the sum of the F_1 scores between the matched communites, and the reported score is then averaged over the true communites. If a method finds less communities than there are true communities, those true communities have 0 F_1 score. Full details are provided in the online appendix.

4.3 Experiments on Synthetic Networks

4.3.1 Synthetic Networks. We generated synthetic networks with a target size of $n = 1500$ vertices. To this end, community sizes for the networks were sampled uniformly from the interval $[50, 150)$. Following the definitions in Subsection 2.1, the communities were generated using the hyperbolic core-tail model, with overlap introduced by allowing their vertex ranges to intersect. To evaluate robustness, we added symmetric random noise by independently flipping edges and non-edges with probability $p_{\text{noise}} \in \{0.0, 0.01, 0.02, 0.04, 0.08\}$. For each p_{noise}, we generated 10 independent network realizations.

Table 1: Number of true and predicted communities.

Alg.	YT	Friend	DBLP	LJ	Amazon	Orkut
# com.	234	1567	47	502	48	4242
HyCSA	234	1567	47	502	48	4242
Louvain	28	139	92	208	63	48
HyCom	168	8047	48	60	15	23618

4.3.2 Parameters. HyCSA was run with a batch size of $B = 10$, an initial temperature T_0 of 0.001, an addition probability of $p_{\text{add}} = 0.6$, and a temperature adjustment parameter of $\alpha = 1$. We evaluated the algorithm using both seeding strategies described in Section 3. The parameter settings used in the synthetic experiments are summarized in the first column of Table 2 in Appendix.

4.3.3 Numerical Results and Discussion: Figures 2a and 2b illustrate the F_1^t Max and F_1^t Mat scores of the algorithms under varying noise levels. HyCSA_{rw} is consistently close to $\text{HyCSA}_{\text{rss}}$, showing that the random walk seed selection performs well in synthetic data sets. Overall, both versions of HyCSA are the best methods, and with higher levels of noise, by a wide margin. Louvain, in particular, suffers from F_1^t Mat (as opposed to Max), as it tends to find much less communities than what there are.

Sub-figure 2c illustrates the *negative log-likelihood of the graph* $-\ell(C \mid G)$ obtained from the synthetic networks across different noise levels (cf. [13, § 4]). Here, the methods are very comparable over different noise levels, indicating that while Louvain and HyCom cannot find the correct communites, they still can describe the data well.

4.4 Experiments on Real-World Networks

In this subsection, we evaluate HyCSA on real-world networks.

Table 2: F_1 scores for algorithms across datasets.

Algorithm	YT	Friend.	DBLP	LJ	Amazon	Orkut
F_1^t Max						
HyCSA$_{\text{rss}}$	0.21	0.75	0.29	0.76	0.93	0.52
HyCSA$_{\text{rw}}$	0.17	0.73	0.18	0.78	0.93	0.48
Louvain	0.16	0.14	0.61	0.66	0.91	0.03
HyCom	0.14	0.68	0.09	0.20	0.16	0.33
F_1^t Mat						
HyCSA$_{\text{rss}}$	0.19	0.75	0.26	0.72	0.93	0.46
HyCSA$_{\text{rw}}$	0.14	0.70	0.17	0.65	0.92	0.39
Louvain	0.05	0.05	0.61	0.37	0.83	0.01
HyCom	0.09	0.67	0.08	0.08	0.16	0.26

Table 3: $-\ell(C \mid G)$ scores for algorithms across datasets.

Algorithm	YT	Friend.	DBLP	LJ	Amazon	Orkut
True comm.	1.536	24.293	0.655	3.046	0.089	212.442
HyCSA$_{\text{rss}}$	1.605	31.355	0.789	4.289	0.099	275.229
HyCSA$_{\text{rw}}$	1.665	32.623	0.813	4.407	0.098	278.620
Louvain	1.380	39.211	0.571	3.889	0.078	280.810
HyCom	1.746	27.900	0.869	7.757	0.169	266.497

4.4.1 Real-World Networks. We use six datasets from the Stanford Network Analysis Project [10]. These datasets provide ground-truth community information, enabling quantitative evaluation of the detected communities using the proposed measures. We can describe the networks briefly as follows [10]:

Youtube (YT) is a social network in which users establish friendship connections and can create groups that other users may join based on shared interests in video content. Friendster is an online gaming network that allows users form a group which other members can then join games. DBLP is a co-authorship network in computer science research papers. Two authors are connected if they publish at least one paper together. LiveJournal (LJ) and Orkut are social networking platforms where users establish friendship connections and create groups that other users can join. Amazon was collected by crawling Amazon website. In the resulting network, two products i and j are connected by an undirected edge if they are frequently purchased together.

From these networks, we retain only communities containing between 50 and 1000 nodes. Number of communities of the resulting networks are summarized in Table 1. More information can be found in the online appendix.

4.4.2 Parameters. HyCSA was run with a batch size of $B = 10$, an initial temperature T_0 of 0.001, an addition probability of $p_{\text{add}} = 0.8$, and a temperature adjustment parameter of $\alpha = 10$. We evaluated the algorithm using both seeding strategies described in Section 3. The parameter settings used in experiments with the real-world networks are summarized in the second column of Table 2 in the online appendix.

4.4.3 Numerical Results and Discussion: Table 2 summarizes the F_1^t Max and F_1^t Mat scores of the algorithms on the real-world networks. Overall, HyCSA achieves competitive performance across the networks. HyCSA$_{\text{rss}}$ is better than HyCSA$_{\text{rw}}$ (F_1^t Max on LJ being an exception), as can be expected. HyCSA$_{\text{rw}}$ is still better than Louvain or HyCom in every data set except Louvain in DBLP. The strong results of Louvain in that data is probably explained by the very strong clique-like structure of the communities (cf. [13]). On the other hand, the reason why HyCSA$_{\text{rw}}$ performs so much worse than HyCSA$_{\text{rss}}$ does need further study. It should also be noticed that with this data, Louvain finds significantly more communities than there are ground-truth communities. It is also worth noticing that HyCom beats Louvain in Friendster and and Orkut data sets, but is significantly worse in other data sets. As Louvain is correspondingly bad in these data sets, we consider our selection of comparative methods to be fair in that Louvain and HyCom are good with different types of data.

Table 3 presents $-\ell(C \mid G)$ scores obtained from the real-world networks as well as the log-likelihood obtained using the ground truth communities. As was the case with the synthetic data, here the methods are more even. HyCSA is no more consistently the best method, but neither is Louvain or HyCom. Yet, HyCSA$_{\text{rw}}$ is always better than then worse of Louvain or HyCom, showing that it has a consistent performance over all data sets. Between HyCSA$_{\text{rss}}$ and HyCSA$_{\text{rw}}$, the differences are typically small; with Amazon data, HyCSA$_{\text{rw}}$ is actually slightly better.

5 Conclusions

Finding pontially overlapping communities is a hard problem. When the communities can have more complex shapes than (quasi-)cliques, the problem becomes even harder as we now need to also model the shape of the community. HyCSA can find such communities reliably from synthetic and real-world data. In some data sets, simpler methods, such as Louvain, can do well, but unlike Louvain or HyCom, HyCSA was the most consistent performer. Tuning the parameters in seeding and simulated annealing for each data set would probably have improved it even more, but we considered such per data set tuning unfair, and used consistent parameter settings accross all data sets. Improving HyCSA on the the data sets it currently struggles the most is left for future work. Future work should also study methods to select the number of communities automatically instead of requiring the user to present it as a parameter.

6 Ethical Considerations

This study evaluates community detection methods using synthetic and publicly available real-world networks from Stanford Network Analysis Project [10]. These real-world networks are widely used for academic research and do not contain personally identifiable or sensitive information. Moreover, our experiments focus on graph structure and community memberships, without attempting to identify individuals or detecting sensitive attributes. Nevertheless, community detection may reveal sensitive associations between individuals, raising potential privacy concerns. Therefore, applications of the proposed method to sensitive real-world data should incorporate appropriate privacy safeguards and ethical considerations.

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